Differentiation

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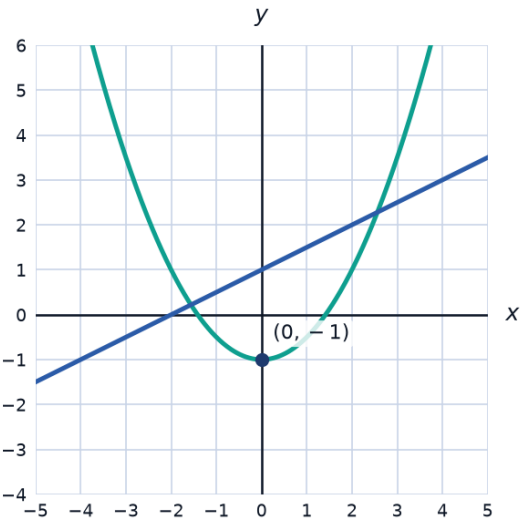
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The chain rule

Find $\frac{dy}{dx}$ for $y = (3x + 1)^5$.

2 [Foundational] • Target: The chain rule

Find $\frac{dy}{dx}$ for $y = \sqrt{4x^2 + 1}$.



3 [Foundational] • Target: The chain rule

Find $\frac{dy}{dx}$ for $y = (2x - 5)^6$.

4 [Foundational] • Target: The product rule

Find $\frac{dy}{dx}$ for $y = x^2 \sin x$.

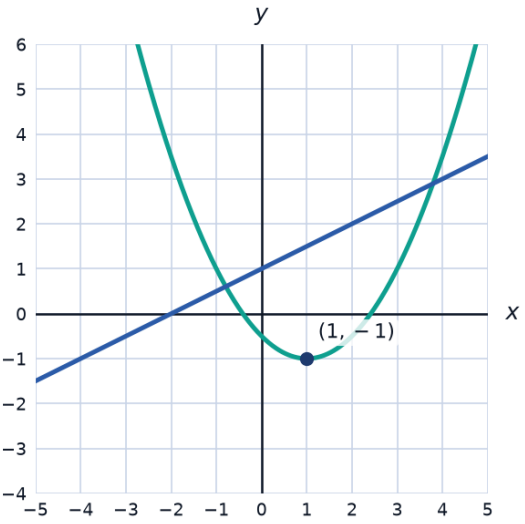
5 [Foundational] • Target: The product rule

Find $\frac{dy}{dx}$ for $y = e^{3x} \cos x$.

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6 [Foundational] • Target: The quotient rule

Find $\frac{dy}{dx}$ for $y = \frac{x^2}{2x+1}$.



7 [Foundational] • Target: The quotient rule

Find $\frac{dy}{dx}$ for $y = \frac{\sin x}{x}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The quotient rule

Find $\frac{dy}{dx}$ for $y = \frac{e^x}{x^2+1}$.

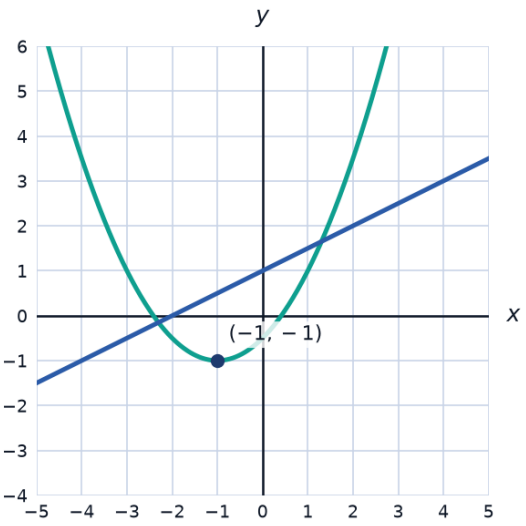
9 [Intermediate] • Target: Implicit differentiation

Find $\frac{dy}{dx}$ given $x^2 + y^2 = 25$.

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10 [Intermediate] • Target: Implicit differentiation

Find $\frac{dy}{dx}$ given $x^3 + y^3 = 6xy$, and evaluate it at the point $(3, 3)$.



11 [Intermediate] • Target: Implicit differentiation

Find $\frac{dy}{dx}$ given $y^2 - xy + x^2 = 7$, and evaluate it at the point $(1, 3)$.

12 [Intermediate] • Target: Implicit differentiation

Find $\frac{dy}{dx}$ given $\sin(xy) = x$, using implicit differentiation.

13 [Intermediate] • Target: Parametric differentiation

A curve is defined parametrically by $x = t^2$, $y = 2t^3$. Find $\frac{dy}{dx}$ in terms of t.

14 [Intermediate] • Target: Parametric differentiation

A curve is defined by $x = 3\cos t$, $y = 3\sin t$. Find $\frac{dy}{dx}$ in terms of t, and find the gradient at $t = \frac{\pi}{4}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

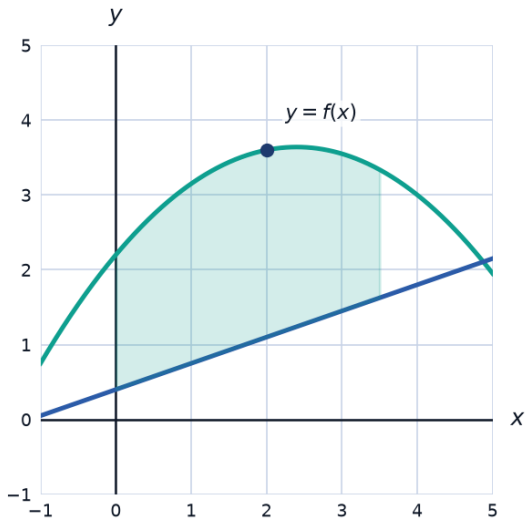
15 [Challenging] • Target: Related rates • Cross-Lesson: 5.01 Integration

A spherical balloon is inflated so its volume increases at a constant rate of 100 cm^3/s. Find the rate of increase of the radius when $r = 5$ cm, using $V = \frac{4}{3}\pi r^3$, to three decimal places.

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16 [Challenging] • Target: Related rates • Cross-Lesson: 6.01 Sequences & Series

The area of a circular ripple increases at 8π cm^2/s. Find the rate of change of the radius when $r = 4$ cm, using $A = \pi r^2$.



17 [Challenging] • Target: Related rates • Cross-Lesson: 7.01 Vectors

A cone-shaped tank (vertex down) has height 10 cm and top radius 4 cm, so at water depth h the radius is $r = \frac{2h}{5}$. Water is poured in at 12 cm^3/s. Find the rate of rise of the water level when $h = 5$ cm, using $V = \frac{1}{3}\pi r^2 h$.

18 [Challenging] • Target: Related rates • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

The radius of a circular oil slick increases at a constant rate of 0.5 m/min. Find the rate of increase of its area when $r = 20$ m.

19 [Challenging] • Target: Differentiating trigonometric, exponential and logarithmic composites • Cross-Lesson: 9.01 Complex Numbers

Find $\frac{dy}{dx}$ for $y = \sin(3x^2)$.

20 [Challenging] • Target: Differentiating trigonometric, exponential and logarithmic composites • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Find $\frac{dy}{dx}$ for $y = \ln(5x^2 + 1)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Product & Quotient Rules

Differentiate $y = x^3e^{2x}$ with respect to x.

a. $x^2e^{2x}(3 + 2x)$

b. $3x^2e^{2x}$

c. $2x^3e^{2x}$

d. $x^2e^{2x}(3x + 2)$

22 [Foundational] • Target: Product & Quotient Rules

Differentiate $y = \frac{\ln x}{x^2}$ for $x > 0$.

a. $\frac{1 - 2\ln x}{x^3}$

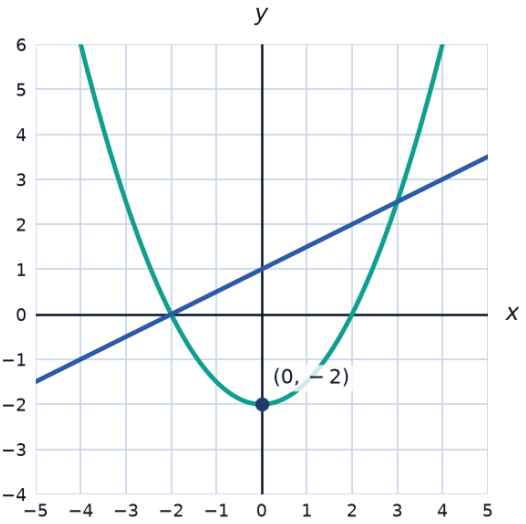
b. $\frac{1 + 2\ln x}{x^3}$

c. $\frac{1 - 2\ln x}{x^4}$

d. $\frac{1}{x^3}$

23 [Foundational] • Target: Product & Quotient Rules

Find the gradient of the curve $y = x\sin(2x)$ at $x = \pi/4$.



a. 1

b. 0

c. $\pi/2$

d. 2

24 [Foundational] • Target: Derivatives of Exponential, Logarithmic & Trig Functions

Differentiate $f(x) = \ln(\sec x + \tan x)$.

a. $\sec x$

b. $\tan x$

c. $\sec^2 x$

d. $\sec x \tan x$

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25 [Foundational] • Target: Derivatives of Exponential, Logarithmic & Trig Functions

Differentiate $y = e^{\sin(3x)}$ with respect to x.

a. $3\cos(3x)e^{\sin(3x)}$

b. $\cos(3x)e^{\sin(3x)}$

c. $3e^{\sin(3x)}$

d. $-3\cos(3x)e^{\sin(3x)}$

26 [Foundational] • Target: Derivatives of Exponential, Logarithmic & Trig Functions

Find $\frac{d}{dx}[\tan^{-1}(2x)]$.

a. $\frac{2}{1+4x^2}$

b. $\frac{1}{1+4x^2}$

c. $\frac{2}{1+2x^2}$

d. $\frac{2}{\sqrt{1-4x^2}}$

27 [Foundational] • Target: Implicit Differentiation

Find $\frac{dy}{dx}$ for the curve $x^2 + xy + y^2 = 7$.

a. $-\frac{2x+y}{x+2y}$

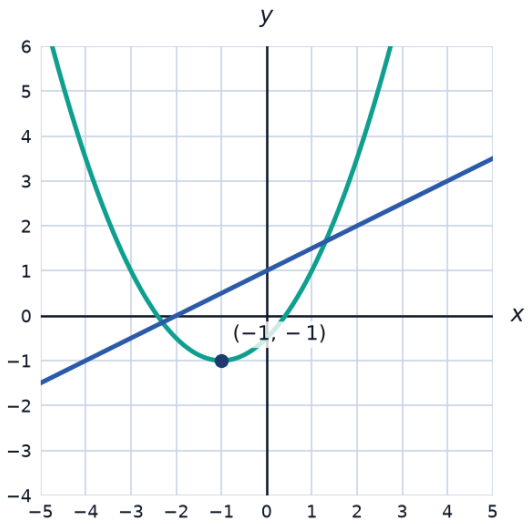
b. $\frac{2x+y}{x+2y}$

c. $-\frac{2x}{2y}$

d. $-\frac{x+2y}{2x+y}$

28 [Foundational] • Target: Implicit Differentiation

Find the gradient of the curve $x^3 + y^3 = 9xy$ at the point $(2, 4)$.



a. $\frac{4}{5}$

b. $-\frac{4}{5}$

c. $\frac{5}{4}$

d. 1

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Implicit Differentiation**

Find the equation of the normal to the curve $x^2 - 2xy + 3y^2 = 9$ at $(3, 2)$.

a. $y = 2$

b. $x = 3$

c. $y = x - 1$

d. $2x + y = 8$

30 **[Intermediate]** • **Target: Parametric Differentiation**

A curve is defined parametrically by $x = 2t^2$, $y = 4t$. Find $\frac{dy}{dx}$ in terms of t.

a. $\frac{1}{t}$

b. t

c. $\frac{2}{t}$

d. $2t$

31 **[Intermediate]** • **Target: Parametric Differentiation**

A curve has parametric equations $x = a\cos\theta$, $y = b\sin\theta$. Find the equation of the tangent at parameter θ.

a. $\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1$

b. $\frac{x}{a}\sin\theta + \frac{y}{b}\cos\theta = 1$

c. $\frac{x}{a}\cos\theta - \frac{y}{b}\sin\theta = 1$

d. $bx\cos\theta + ay\sin\theta = ab$

32 **[Intermediate]** • **Target: Parametric Differentiation**

Find $\frac{d^2y}{dx^2}$ for the parametric curve $x = t^2$, $y = t^3$.

a. $\frac{3}{4t}$

b. $\frac{3}{2}$

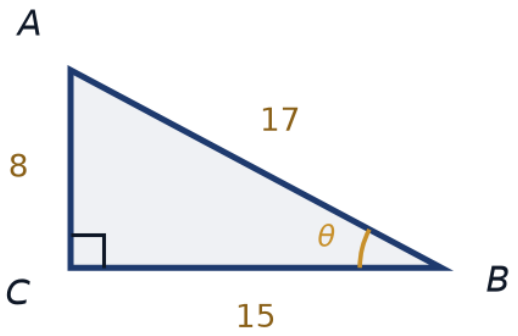
c. $\frac{3}{4t^2}$

d. $\frac{6}{t}$

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33 [Intermediate] • Target: Parametric Differentiation

A straight line passing through the origin is tangent to both the curve C_1 with equation $y = e^{kx}$, where k is a positive constant, and the curve C_2 with equation $y = \ln x$. Determine the exact distance between the two points of contact.



a. $(e-1)\sqrt{e^2+1}$

b. $e\sqrt{e^2+1}$

c. $(e-1)\sqrt{e^2-1}$

d. $\sqrt{e^4-e^2+1}$

34 [Intermediate] • Target: The chain rule

The first, fourth, and sixth terms of an arithmetic progression (u_n) are equal to the first, second, and third terms, respectively, of an infinite geometric progression (v_n) with common ratio $re1$. Given that the sum to infinity of the geometric progression is 27, determine the maximum value of $S_n = \sum_{k=1}^{n} u_k$, the sum of the first n terms of the arithmetic progression.

a. 44

b. 45

c. 50

d. 54

35 [Intermediate] • Target: The product rule

The function f is defined for $x \in \mathbb{R}$ by $f(x) = 3x + k$, where k is a constant. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x-4|$. Given that $f^{-1}(4) = 2$, find the value of k and all real solutions to the equation $f^{-1}(x) = g(x)$.

a. $k = -2;\ x = \frac{5}{2},\ x = 7$

b. $k = -2;\ x = 7$

c. $k = 2;\ x = \frac{7}{2},\ x = 5$

d. $k = -2;\ x = -\frac{5}{2},\ x = 7$

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36 [Intermediate] • Target: The quotient rule

The curve C has equation $y = \frac{ax+6}{x^2+3}$ where a is a constant. Given that C has a stationary point at $x = 1$, find the coordinates of the other stationary point of C.

a. $(-3, -1)$

b. $(-3, 1)$

c. $(3, 1)$

d. $(-3, -2)$

37 [Intermediate] • Target: Implicit differentiation

A curve has equation $y = \ln(2x^3 + 1)$. What is the gradient of the tangent to the curve at the point where $x = 1$?

a. 2

b. 6

c. $\frac{2}{3}$

d. $\ln 3$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Parametric differentiation • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

What are the values of x that satisfy the equation $|3x - 2| = 2|x + 1|$?

a. $x = 0$ and $x = 4$

b. $x = 0$ and $x = -4$

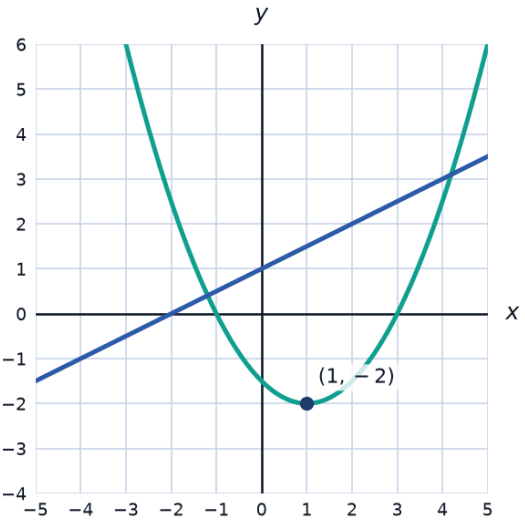
c. $x = \frac{1}{5}$ and $x = 4$

d. $x = \frac{1}{4}$ and $x = \frac{3}{2}$

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39 [Challenging] • Target: Related rates • Cross-Lesson: 9.01 Complex Numbers

A curve has equation $y = x^2e^{3x}$. Find the gradient of the curve at the point where $x = 1$.



a. $5e^3$

b. $6e^3$

c. $3e^3$

d. $2e^3$

40 [Challenging] • Target: Differentiating trigonometric, exponential and logarithmic composites • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Given $f(x) = -3x + 4$ and $g(x) = -4x - 6$, find $gf(3)$.

a. 14

b. 58

c. -18

d. -5

41 [Challenging] • Target: Applying the rules: tangents and second derivatives • Cross-Lesson: 10.01 Numerical Methods

The function $f(x) = (x + 7)^2 - 15$ has domain $x \geq -7$. State the range of f.

a. $f(x) \geq -7$

b. $x \geq -7$

c. $f(x) \leq -15$

d. $f(x) \geq -15$

42 [Challenging] • Target: A well-designed word problem: a ladder sliding down a wall • Cross-Lesson: 11.01 Differential Equations

Given $f(x) = -5x - 18$, evaluate $f(|0|)$.

a. -15

b. -18

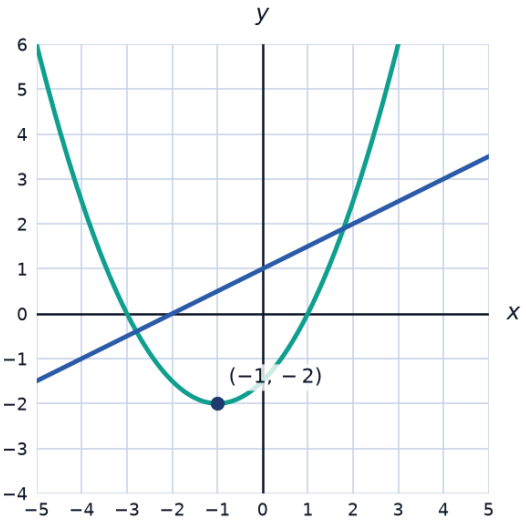
c. -14

d. 18

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43 [Challenging] • Target: Product & Quotient Rules • Cross-Lesson: 12.01 Poisson Distribution

Given $f(x) = -x - 10$ and $g(x) = -3x + 9$, find $fg(6)$.



a. -1

b. 57

c. -9

d. -16

44 [Challenging] • Target: Derivatives of Exponential, Logarithmic & Trig Functions • Cross-Lesson: 13.01 Continuous Random Variables

The function $f(x) = (x + 11)^2 + 5$ has domain $x \geq -11$. State the range of f.

a. $x \geq -11$

b. $f(x) \geq -11$

c. $f(x) \leq 5$

d. $f(x) \geq 5$

45 [Challenging] • Target: Implicit Differentiation • Cross-Lesson: 14.01 Sampling & Estimation

The function $f(x) = (x + 6)^2 - 3$ has domain $x \geq -6$. State the range of f.

a. $f(x) \leq -3$

b. $f(x) \geq -3$

c. $f(x) \geq -6$

d. $x \geq -6$

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